Software Requirements Specification (SRS)

Protected Premise Security Monitoring System

Version 1.0
Based on IEEE Std 830-1998

1. Introduction

1.1 Purpose

The purpose of this document is to specify the software requirements for the Security System Software . This system is designed to monitor and log the entry/exit of personnel and assets within a restricted environment. The intended audience includes the project development team, project stakeholders, and system evaluators.

1.2 Scope

The Using the IEEE 830-1998 standard as a guideline, develop a software requirements specification (SRS) document.

Notes:

1. See attached Std-830-1998-software-requirements-specifications_SRS.pdf document

2. Use an appropriate template to structure your SRS document (for example: refer to section A7 on pages 24 and 25 of the attached standard 830-1998 document.

software will provide a centralized interface for premises security. It involves real-time tracking of individuals via access badges/cameras and the inventory tracking of electronic items using digital manifests. The system will alert security personnel of unauthorized entry or asset removal.

1.3 Definitions, Acronyms, and Abbreviations

PPSMS: Protected Premise Security Monitoring System.
SRS: Software Requirements Specification.
Premise: The physical restricted area under surveillance.

1.4 References

1. IEEE Std 830-1998, IEEE Recommended Practice for Software Requirements Specifications.
2. Project Schedule and Gantt Chart (Version 2.0).

1.5 Overview

Section 2 provides a general description of the system's functions and users. Section 3 detail the specific functional and non-functional requirements.

2. Overall Description

2.1 Product Perspective

This system is a stand-alone security solution but provides API hooks for integration with existing institutional databases for personnel verification.

2.2 Product Functions

• Real-time monitoring of entry/exit points.
• Automated logging of personnel IDs and timestamps.
• Asset tracking and verification (matching items to owners).
• Real-time security alerts/notifications.

2.3 User Characteristics

• Security Officers: Primary operators with monitor and alert intervention rights.
• System Admin: Responsible for database maintenance and user access management.

2.4 Constraints

The system must operate 24/7. Data logs must be kept for a minimum of 6 months. Hardware integration is limited to standard IP-based cameras and RFID readers.

2.5 Assumptions and Dependencies

It is assumed that the premise has a stable internal network (LAN) and that personnel will comply with scanning protocols.

3. Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The dashboard shall display a live list of the last 10 entries/exits and a graphical overview of total people currently inside the premise.

3.2 Functional Requirements

3.2.1 Entry Monitoring

The system shall record the unique ID of every person entering through valid access points.

3.2.2 Exit Validation

The system shall cross-reference outbound assets with the entry manifest to ensure no unauthorized items are leaving the premise.

3.3 Performance Requirements

The system shall process scan data and update the display within 0.5 seconds of the scanning event.

3.4 Design Constraints

The backend must be developed using a relational database (SQL) for reliability and data integrity.

3.5 Software System Attributes

3.5.1 Security

Access to the log database shall be restricted using 256-bit encryption and multi-factor authentication for administrators.